Redhill & Shoreham Aviation - Piper PA28 Checklist

Name: _____

31st March 2021 - Revision 1.0

External Checks

- Check documents are current and correct.
- Visually check the aircraft's general condition.
- Remove tie-down ropes, pitot head cover, chocks and any external control locks.
- In cold weather, remove even small accumulations of frost, ice or snow from the aircraft.
- Check that the aircraft is parked in a free area, so that it may easily move onto the taxyway.
- Check that a fire extinguisher is available.

1. Cockpit Preparation

- Aircraft Documents.....Stowed
- Parking Brake.....On
- I.F. Screens.....Stowed
- First Aid Kit.....Stowed
- Fire Extinguisher..... Checked
- Magneto Switches.....Off
- Avionics Master Switch.....Off
- Circuit Breakers..... Check In
- Trim Indicator.....Correct Movement/Set
- Flying Controls.....Full Free Correct Movement
- Battery Switch.....On
- Pitot Heat.....On
- Fuel Contents..... Check
- Fuel selector.....Emptier Tank
- Annunciator/low voltage lights..... Test/On
- *Leave cockpit and check externally as necessary, and then re-enter cockpit before carrying out full external pre-flight inspection.*
- *Recognition/landing lights.....On/Check/Off*

1. Cockpit Preparation (Continued)

- *Anti-collision lights .....On/Check/Off*
- *Navigation lights.....On/Check/Off (Leave on at night)*
- *Pitot heater.....On/Check/Off*
- *Stall Warner.....On/Check/Off*
- *Panel lighting..... Check (Leave on at night)*
- Battery switch.....Off
- Pitot static drains.....Operate
- Flaps.....Fully Down/Check All Stages

2. External: Cabin Upper Fuselage

- Door.....Closed Flush
- Starboard Windows..... Clean
- Upper Fuselage.....Check/Undamaged
- Towbar (If Carried).....Stowed
- Baggage Door..... Secure

3. Starboard Wing

- Flap.....Check/Undamaged
- Aileron (Surface, Hinges & Static Wicks).....Check/ Undamaged
- Wing Tip..... Undamaged
- Strobe & Navigation Light..... Undamaged
- Wing Surfaces..... Undamaged
- Fuel Tank (Contents, Cap Secure, Vent, Drain, No Seepage)..... Check
- Main Gear (Oleo Extension 4½")..... Check Security
- Tyre (Inflation, Creep, No Damage)..... Check
- Brakes (Pads, No Hydraulic Leaks, Undamaged)..... Check
- Fresh Air Intake..... Unobstructed

4. Nose Section

- Windscreen..... Clean
- OAT Probe..... Check
- Oil Contents (8 Qts Min 2 Qts, Recommended Minimum 6 Qts)..... Check/Cap Secure
- Starboard Cowling..... Secure
- Propeller/Spinner.....Undamaged/Secure
- Air Intakes..... Unobstructed
- Alternator Belt..... Check Tension

4. Nose Section (Continued)

- Landing Light..... Undamaged
- Nose Gear (Oleo Extension 3¼")..... Check Security
- Tyre (Inflation, Creep, No Damage)..... Check
- Brake Fluid..... Check
- Port Cowling..... Secure
- Fuel Drain..... Sample/No Seepage
- Port Window..... Clean

5. Port Wing

- Fresh Air Intake..... Unobstructed
- Brakes (Pads, No Hydraulic Leaks, Undamaged)..... Check
- Tyre (Inflation, Creep, No Damage)..... Check
- Main Gear (Oleo Extension 4½")..... Check Security
- Fuel Tank (Contents, Cap Secure, Vent, Drain, No Seepage)..... Check
- Pitot Head/Static.....Unobstructed/Secure
- Stall Warning Vane.....Free/Undamaged
- Wing Surfaces..... Undamaged
- Wing Tip..... Undamaged
- Strobe & Navigation Light..... Undamaged
- Aileron (Surface, Hinges & Static Wicks).....Check/ Undamaged
- Flap.....Check/Undamaged

6. Rear Fuselage, Fin And Stabilator

- Radio Aerials..... Secure
- Port Fuselage & Window.....Undamaged/Clean
- Fresh Air Intake..... Check

- Fin Surface.....Undamaged/Secure
- Rudder & Nav Light..... Undamaged
- Stabilator.....Free/Undamaged
- Air Conditioning Scoop (If Fitted)..... Flush/Undamaged
- Starboard Fuselage & Window.....Undamaged/Clean
- Static Vents (If Fitted).....Unobstructed/Clean

Before Engine Start Checks

1. Flaps.....Retracted
2. External Checks.....Completed
3. Passenger Briefing.....Completed
4. Cabin Door.....Closed & Latched
5. Seats.....Adjusted & Locked
   - Seat Backs Upright
6. Seat Belts & Shoulder Straps.....Fitted & Secure
7. Alternate Static Selector (If Fitted).....Check/Normal
8. Flight Instruments.....Check Condition
   - Standby Compass & Deviation Card
9. Battery Master Switch.....On
10. Annunciator/Low Voltage Light.....Check/On
11. VAC Off Caption.....On
12. Avionics/Radio Master Switch.....Off

Starting

1. Anti-Collision Light.....On
2. Fuel Selector.....Tank With Lowest Contents
3. Mixture.....Exercise/Set Rich
4. Throttle.....Exercise/Set ¼”
5. Electric Fuel Pump.....On, check pressure, Off
6. Prime.....As Required & Lock
7. Lookout.....Clear All Round, Open Window “CLEAR PROP”
8. Magnetos.....On Both
9. Starter.....Operate & Release as Soon as Engine Starts

After Start

1. Starter Warning Light.....Out, If Not Immediately Turn Magnetos Off
2. Oil Pressure.....Within 30 secs 25 PSI Minimum (If Not, Shut down)
3. RPM.....Set to 1200
4. Alternator/Ammeter.....On & Holding Positive Charge
5. Low Voltage Light.....Out
6. Suction.....Check Registering, VAC off Caption Out
7. Annunciator Panel Lights.....All Out
8. Fuel Pressure.....Check
9. Avionics/Radio Master Switch.....On (If Fitted)

10. Electric Trim (If Fitted).....Check
11. Flight Instruments.....Check (AI erect, DI set, VSI, Turn Co-ord flag away)
    - HSI/RMI Heading, Standby Compass/Slaving Meter
12. Avionics.....Checked and Set As Required
13. Auto-Pilot (If Fitted).....Check and Off
14. Magnetos.....Check at 1200 RPM
15. Radios.....Radio Check, ATIS (if available) & Taxi Clearance
16. Altimeters.....Cross Check & Set

Taxying

1. Brakes & Nosewheel Steering.....Check Operation Satisfactory For Both Pilots
2. Flight Intruments.....During Turns Check Compass & D.I turning in same
    - direction, turn Indicator: Turn & Skid. AI steady

Power Checks

1. Position.....Into wind. Avoid Rough Ground and Loose Stones.
    - Clear of Aircraft and Personnel
2. Parking Brake.....On
3. Fuel Selector.....Change Tank to Fuller Tank
4. RPM.....Check Clear All Round. Set to 2000
5. Carb. Heat.....Check Operation & Return to Cold
6. Magnetos.....Check Each Magneto in Turn. Max Permissible Drop
    - 175 RPM; and Max. Differential is 50 RPM
7. Alternator Output.....Check
8. Suction.....4.9 to 5.1” Hg.
9. Temperature & Pressure.....Check, Steady and Within Limits
10. RPM.....Smoothly Close the Throttle, Check Idle.Temps and
    - Pressures Steady, Within Limits. Then Reset 1200

Pre-Takeoff Checks

1. Trim.....Set for Take-Off
2. Throttle Friction.....Finger Tight
3. Mixture.....Mixture Rich.
4. Magnetos.....Magnetos on Both. Master Switch On
5. Carb Heat.....Off
6. Fuel Selector.....On Fuller Tank, Check Contents
7. Fuel Pump.....On, Pressure Checked
8. Flaps.....Set (Normal 0°)
9. Engine Instruments.....Check (Temperatures & Pressures)
10. Seat Belts / Seat Backs.....Secure/Erect
11. Cabin Doors.....Locked/Latched
12. Annunciator Panel.....Test/All Lights Out
13. Flying Controls.....Full Free Correct Movement

14. Radio & Transponder.....Confirm Radio/Nav Set. Transponder set, Alt.
15. Flight Instruments.....Check/Altimeters Set. Pitot Heat as Required
16. Lights.....Recognition/Landing Lights and Anit-Cols Lights On
17. Take-Off Brief.....Review Speeds, Crosswind, Departure Brief &
    - Emergency Procedures

Take Off

Normal Take-Off

1. Lookout.....Obtain Clearance, Check Approach Clear Visually
2. Line up.....On Runway Centreline, Ensure Nosewheel Straight,
    - D.I. Indicating Runway Heading. Controls Free
3. Brakes.....Off, Heels on Floor
4. Throttle.....Smoothly Open to Full; Check Power
5. Oil.....Temperature and Pressure Steady and in Green
6. Airspeed.....Increasing
7. Elevator Control.....Lift Nose by 60 Kts
8. Climb Speed.....80 Kts (Clean).

Short Field Take-Off (Hard Surface)

1. Flaps.....25°
2. Brakes.....Hold (Keep Moving During a Soft Field Take-Off)
3. Throttle.....Fully Open, Check power, Oil Temp and Pressure
4. Brakes.....Release
5. Elevator Control.....Slight Back Pressure
6. Climb Speed.....80 Kts (Best angle of Climb is 65 Kts)

After Take-Off

1. Lookout.....Point Selected Left of Nose, Check Drift
2. Flaps.....Retract Above 200' a.g.l, Not Below 65 Kts
3. Oil.....Temperature and Pressure Steady, Within Limits
4. Recognition/Landing Lights.....Off
5. Fuel Pump.....Off (Above 1000ft a.g.l)

Climb

Normal Climb

1. Airspeed.....Clean: 80 Kts (Best Rate of Climb)
2. Throttle.....Fully Open
3. Mixture.....Rich Below 3000'

En-Route Climb

1. Airspeed.....90 Kts
2. Throttle.....Fully Open
3. Mixture.....As Required

En-Route Checks (FREDA)

1. **F**uel.....Check Contents/Balanced
2. **R**adio / Radio aids.....Check Volume; Frequency Change if Required /
   - Radio Aids Identified
3. **E**ngine.....Oil Temperature and Pressure Steady and Within
   - Limits. Ammeter Charging. Suction Within Limits.
   - Check for Carb. Icing. Mixture as Required.
4. **D**irection Indicator.....Check D.I.
5. **A**ltimeter.....Set

HASELL Checks (Stalls)

1. **H**eight.....Sufficient to Recover by 2000’ AGL
2. **A**irframe.....Flaps as Required. Brakes off
3. **S**ecurity.....Doors, Windows and Harnesses Secure; No Loose
   - Articles
4. **E**ngine.....Check Oil Temperature and Pressure, Mixture as
   - Required, Check Fuel Contents, Fuel Pump On,
   - Clear Carb Ice
5. **L**ocation.....Clear of Airfields, Built Up Areas, Controlled
   - Airspace or Cloud and Danger Areas
6. **L**ookout.....Check No Aircraft Around, Particularly Below

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HELL (For Continued Exercise)

1. **H**eight.....Sufficient to Recover by 2000’ AGL
2. **E**ngine.....Check Oil Temperature and Pressure, Mixture as
   - Required, Check Fuel Contents, Clear Carb Ice
3. **L**ocation.....Clear of Airfields, Built Up Areas, Controlled
   - Airspace or Cloud and Danger Areas
4. **L**ookout.....Check No Aircraft Around, Particularly Below

Note: Reset D.I after completion of the exercise

Airfield Approach Checks: FFREDA

1. **F**uel.....Check Quantity/Balanced
2. **F**uel Pump.....On
3. **R**adio.....Correct Frequency Volume Up. Radio Aids Identified
4. **E**ngine.....Temperature and Pressure in Limits, Mixture Rich
5. **D**irection Indicator.....D.I. Reset
6. **A**ltimeters.....Altimeters Set

Pre-Landing Checks

1. Brakes.....Brakes Off; Pressure Up
2. Mixture.....Mixture Rich
3. Carb. Heat.....Carb. Heat, Check for Icing and then Return to Cold
4. Fuel.....Fuel On, Sufficient for Go Around
5. Oil.....Oil Temperature and Pressure Within Limits
6. Hatches and Harnesses.....Pilot and Passenger Secure
7. Lights.....Recognition/Landing Lights On

Approach and Landing

Normal Landing

1. Carb. Heat.....Apply Full Heat Before Closing Throttle
2. Flaps.....Select 25° When Speed is Below VFE
3. Airspeed.....Initially 70 Kts. Final Approach, 63 Kts, Flap 40°
4. Touchdown.....Mainwheels First, Lower Nose Wheel Gently
5. Braking.....Minimum Required

Flapless Landing

1. Airspeed.....75 Kts

Shortfield Landing

1. Flaps.....40°
2. Airspeed.....65 Kts
3. Touchdown.....Main Wheels, Lower the Nose
4. Brakes.....Apply Positively

Glide Approaches

- Airspeed.....75 Kts
- Flap.....As Required

Notes:

1. During prolonged glides; the engine should be warmed every 500’.
2. The approach speeds are for calm conditions, in gusty or turbulent conditions the figures should be increased.

Go-Around (Missed Approach)

1. Throttle.....Full Power
2. Carb. Heat.....Cold
3. Airspeed.....65 Kts Until Clear of Obstacles, then 80 Kts
4. Flaps.....If 30°, Retract to 25° as Soon as Possible. Retract
   - Remaining 25° Above 200’
5. Radio.....Inform ATC

After Landing

Clear the runway and stop the aircraft

1. Carb Heat.....Cold
2. Flaps.....Up
3. Fuel Pump.....Off
4. Pitot heat.....Off
5. Strobe Lights / Landing Lights.....Off / As Required
6. Trimmers.....Set Neutral
7. Transponder.....Standby
8. Non-essential Electrics.....Off

Shut Down

1. Park.....Into Wind, Nosewheel Straight
2. Parking brake.....On
3. Engine.....1200 RPM for 30 secs; Check Oil Temperature and
   - Pressure
4. Magnetos.....Check
5. Avionics/Radio Master Switch.....Off
6. Mixture.....Full Lean
7. Throttle.....Closed

After Engine Stop

1. Magnetos.....Off, Key Out
2. Exterior Lights.....All Off
3. Fuel Selector.....Off
4. Interior Lights.....Off
5. Battery & Alternator Masters.....Off
6. Harnesses.....Tidy
7. External.....Pitot Cover On. Lock up and tie down. Use Chocks
   - if necessary

End of Normal Checklist

Emergencies

Engine Failure After Takeoff

If the engine fails after take-off you should (as soon as you notice the loss of power):

1. Lower the Nose and Maintain the Aircraft's Gliding Speed
2. Close the Throttle
3. Select the Most Suitable Landing Area, Compatible With Obstructions and Wind
   - Direction; Preferably Straight Ahead or Within 30° Either Side
4. Maintain Speed and Use Flap as Necessary
5. If Time Permits: - Mixture to I.C.O., Fuel Off, Fuel Pump Off, Magnetos Off, Door
   - Unlocked and Harnesses Tight, Mayday Call, Master Switch Off
6. After Touchdown Leave the Aircraft as Quickly as Possible

Crash Drill

1. Mixture I.C.O, Fuel Pump Off
2. Fuel Selector Off
3. Magnetos Off
4. Brakes Off
5. Harnesses Very Tight
6. Hatches Unlatched
7. Master Switch Off (After Electrical Services are No Longer Required)

Engine Fire During Start

1. Starter - Crank Engine
2. Mixture - Idle Cut Off
3. Throttle - Open
4. Electric Fuel Pump - Off
5. Fuel Selector - Off
6. Abandon Aircraft if Fire Continues

Engine Fire In The Air

1. Fuel Selector - Off
2. Throttle - Closed
3. Mixture - Idle Cut-Off
4. Electric Fuel Pump - Check Off
5. Heater and Defroster - Off
6. Proceed with Power Off Landing Procedure

Electrical Fire In The Air

1. Master Switch Off, Alternator Off
2. Vents - Open
3. Cabin Heat - Off
4. Land as soon as Practicable.

Cabin Fire In The Air

1. Master Switch Off
2. Vents, Cabin Air/Heat Closed
3. Use Fire Extinguisher and Then Ventilate The Cabin
4. Land as Soon as Possible.

Forced Landing Without Power

1. Adopt Glide and Trim for Correct Gliding Speed
2. Check Wind Direction
3. Select Landing Area and 1000’ Area
4. Plan Circuit
5. Complete Failure Checks:
   - Switch Tanks
   - Electric Fuel Pump On
   - Mixture Rich
   - Carb. Heat Change Position
   - Magnetos on Both
   - Primer Locked
6. Mayday Call on Radio
7. Complete Crash Drill
8. Use Full Flap for the Touchdown if Possible
9. Apply Heavy Braking

Forced Landing With Power

1. Assess the Time Available and Check Fuel State, Hours of Daylight and Rate of
   - Weather Deterioration
2. Check Wind Direction
3. Select Suitable Landing Area
4. Carry Out an Inspection Run at Slow Safe Cruising Speed; Initially at 500’ to Check
   - for Obstructions and Secondly at 100’ Over the Landing Area to Reassess the
   - Approach Path, Landing Surface and Overshoot Area (if time permits)
5. Set up For a Short or Soft Field Landing (as appropriate)

Ditching

1. Set up Minimum Rate of Descent (75 Kts)
2. Mayday Call
3. Secure or Jettison Heavy Objects
4. With High Winds and Heavy Seas: Approach Into Wind.
   - With Light Winds and Heavy Swells: Approach Parallel to Swells.
5. Unlatch the Door
6. Touchdown in a Level Attitude
7. Evacuate Through the Cabin Door; if Necessary Open Window to Equalise Pressure

Notes:

1. Ideally, land ahead and either side of a small ship.
2. Do not inflate life jackets in the cockpit.

Low-Voltage/Electrical Failure

1. Check Ammeter, if Reading Zero, Alternator Off
2. Reset Circuit Breaker
3. Alternator On
4. Reduce Electrical Load

Notes:

1. If total electrical failure occurs, keep an extra good lookout. Land when safe to do so,
   - clear the runway as quickly as possible and phone ATC at once.

Radio Failure

1. Check Correct Frequency
2. Check Squelch Setting (if applicable)
3. Check Volume
4. Check Microphone Plug Secure
5. Check Circuit Breaker
6. Try to Communicate Using the Other PTT Switch

Notes:

1. If the radio is unserviceable: flash the landing light on approach and land on visual
   - signals.
2. If flying outside of the circuit, follow lost comm procedure from AIP.

Uncertain of Position / Lost Procedure

Uncertain of Position

Inform ATC immediately on the frequency you are in communication with. Check flight plan, log, compass and heading for gross errors; calculate DR position; from flight time decide radius of circle of uncertainty; mark on map using DR position as Centre; estimate new heading from DR position to regain destination or track. Utilise Nav aids including moving map if available.

Lost Procedure

a) Radio Available

- Check frequency selection; declare that you are lost; make maximum use of aids
- available; consider increasing height to improve range (taking account of controlled
- airspace). R/T procedure: urgency or distress call to obtain fix. You are no longer lost.
- Follow the ATC instructions in full.

b) Radio Not Available

i. Check remaining fuel and calculate endurance; consider daylight remaining, weather
   - tendency, nature of terrain in relation to probable position, minimum safe altitude; recheck flight log for errors; switch transponder, if available, to emergency code.

ii. Map reading from ground to map and try to identify the likely area from the nature of
   - the terrain before checking other features. Look for prominent features in the distance
   - as well as nearby; adhere to a calculated heading and keep map properly orientated.

iii. Decide on a positive course of action; relative merits of square search, flight towards
   - prominent or line feature outside of circle of uncertainty; consider precautionary
   - landing (see Ex 17) if you are short of fuel or are unable to continue the flight.

End of Emergency Checklist

Altimeter Setting Procedures

No. 1	No. 2	Description
Pre-flight	Check +50 ft / -75 ft of datum, 50 ft of each other	
Airport QFE	Airport QFE	
Airport QNH	Airport QNH	
Departure	For take-off	Airport / zone QNH
	Cleared to a FL.	Airport / zone QNH
	Airport QNH	1013
Cruise	Above Transition Alt	Regional QNH
	At / below Transition Alt	Regional QNH
	1013	Regional QNH
Note	In or below TMA use TMA QNH not regional QNH	
Descent	Cleared to FL	Zone / regional QNH
	Cleared to Altitude	Zone / regional QNH
	1013	Zone / airport QNH
Final Approach / Circuits	Airport QNH / QFE	Airport QNH
Missed Approach	Airport QNH	Airport QNH
Note	Airport QNH is valid within 25m of airport	

Recommended Operating Speeds / Limitations

Category	Archer 181	Warrior 161	Notes
Take-Off			
Rotate (depending on weight)	60	60	Reduced by 5 kts for soft fields
Climb			
0° Flap / Best Rate	79	79	
0° Flap / Best Angle	65	63	
Cruise Climb	90	90	
Minimum speed of 60 kts for flap retraction			
Cruise			
Normal / 2200 RPM	100	90	
Bad Vis. 20° Flap / 2000 RPM	70	70	
Approach			
Normal (25° Flap)	70	70	1. For max performance approach speed is 63 kts
Glide	75	75	2. Add 5kts to take-off speeds and VAT in strong winds / turbulence, reducing flap settings as necessary
Flapless	75	75	
Performance / Short-field (40° Flap)	65	65	
Max Airspeeds			
VFE (White Arc)	102	103	
VNO (Green Arc)	125	126	
VNE (Red Line)	154	160	
VA (Manoeuvring Speed @ MAUW)	113	111	
Max X-Wind	17	17	
Crosswind Angle	10°	20°	30° / 40° / 50° / 60° / 70° / 80° / 90°
Crosswind Factor	0.2	0.3	0.5 / 0.6 / 0.7 / 0.8 / 0.9 / 0.9 / 1.0